

Video Preprocessing Documentation

Objective

Prepare video files for transcription by converting them to MP3 format and processing them with OpenAI Whisper.

Tools Used

- FFmpeg: Extracts audio from video files and converts them into MP3 format.
- OpenAI Whisper: Transcribes MP3 audio files into text.

Implementation

To reduce computational load during transcription, videos are first converted into MP3 files. A Python script named **process_video.py** automates this preprocessing stage by reading all video files from the **video/** directory, converting them into MP3 files using FFmpeg, saving them in the **mp3/** directory, and preparing them for transcription using OpenAI Whisper.

Workflow

video/ → process_video.py → FFmpeg → mp3/ → OpenAI Whisper → Transcript (.txt)

Conclusion

FFmpeg extracts audio from video files by converting them into MP3 format, reducing the computational load for transcription. OpenAI Whisper processes the generated MP3 files to produce accurate text transcripts, which are then used as the input for the Retrieval-Augmented Generation (RAG) pipeline.